

Fm16 LeonMcQueen Proof 3

Review

$$1 \quad e^{3x-1} \ln 2x$$

$$\frac{3e^{3x-1} \ln 2x + \frac{1}{2} e^{3x-1}}{e^{3x-1} (3 \ln 2x + \frac{1}{2})}$$

$$e^2 (3 \ln 2 + \frac{1}{2}) \quad 6$$

$$2 \quad |2x+3| > 9$$

$$\therefore 2x+3 > 9 \quad \text{or} \quad -3-2x > 9$$

↓

$$2x > 6$$

$$-2x > 12$$

$$x > 3$$

$$x < -6$$

$$\therefore x < -6 \quad \text{or} \quad x > 3$$

$$3 \quad 2x - 2x \frac{dy}{dx} - 2y + 3y^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2(x-y)}{2x-3y^2}$$

$$4 \quad f^{-\frac{1}{4} \cos^4 x + c}$$

$$5 \quad \frac{A}{x-1} + \frac{B}{x-2} + \frac{C}{x-3} = \frac{x^2+1}{(x-1)(x-2)(x-3)}$$

$$A(x-2)(x-3) + B(x-1)(x-3) + C(x-1)(x-2) = x^2+1$$

$$A = -\frac{1}{2}$$

$$B = -1$$

$$C = \frac{1}{2}$$

$$\therefore \frac{x^2+1}{(x-1)(x-2)(x-3)} = \frac{1}{2(x-3)} - \frac{1}{x-2} + \frac{1}{2(x-1)}$$

1 A

2 a) All rich people are sad

a) There is at least one prime number between 10^7 and 1.1×10^7

c) if p and q are prime numbers then $pq+1$ is not a prime number.

d) Not all numbers of the form 2^n-1 are either prime or a multiple of 3

e) None of the statements above are true.

3 a) If n^2 is odd, then n is even.

b) Let $2n+1$ be an odd number. Then $(2n+1)^2$ is even.

$$\text{Then } (2n+1)^2 = 4n^2 + 4n + 1$$

$$= 2(2n^2 + 2n) + 1 \quad \text{takes form } 2k+1$$

which cannot be even! $\therefore$ odd $\square$

4 a) let $2k$ be the greatest even integer

$$\text{But } 2k+2 > 2k \\ (2 > 0)$$

And $2k+2 = 2(k+1)$: even

$\therefore 2k$ is not the greatest integer!

$\therefore$ no greatest integer $\square$

6) let $2k$ be even integer. Then our claim is that $(2k)^3$ is odd

$$(2k)^3 = 8k^3 = 2(4k^3): \text{ an even integer.}$$

$\therefore (2k)^3$ cannot be odd.

$\therefore$ if n is even, n^3 is even $\square$

c) let p be an odd number $2n+1$

let q be another odd number $2m+1$

Claim: pq is even.

$$pq = (2n+1)(2m+1)$$

$$= 4mn + 2n + 2m + 1$$

$$= 2(2mn + m + n) + 1 = 2k+1 \quad (k = 2mn + m + n)$$

$2k+1$ is odd! so pq cannot be even.

$\therefore$ if pq is even $2 \mid p$ or $2 \mid q$

d) let p be an even integer $2n$.

let q be an odd integer $2m$

Claim: pq is odd.

$$\text{But } pq = 2n \cdot 2m = 2(nm): \text{ even!}$$

so pq cannot be odd.

so for pq to be odd one of p or q must be odd.

5 i) let $a = \frac{p}{q}$ where $p, q \in \mathbb{Z}$

$$\text{let } b = \frac{x}{y} \text{ where } x, y \in \mathbb{Z}$$

$$\text{Then } a+b = \frac{p}{q} + \frac{x}{y}$$

$$= \frac{py + qx}{qy}$$

$$py \in \mathbb{Z}$$

$$qx \in \mathbb{Z}$$

$$qy \in \mathbb{Z}$$

$a+b \in \mathbb{Q}$ as numerator & denominator
are both integers.

i) Claim: if $a+b \in \mathbb{Q}$, then $a, b \in \mathbb{Q}$.

$$0 \in \mathbb{Q}.$$

$$\text{However, } \sqrt{2} + (-\sqrt{2}) = 0$$

$$\text{But } \sqrt{2} \notin \mathbb{Q}$$

$$-\sqrt{2} \notin \mathbb{Q}$$

$\therefore a, b$ may not be in $\mathbb{Q}$!

$\therefore$ Claim is false $\square$.

ii) Claim: if a is rational and b is irrational, then $a+b$ is rational.

$$\text{Let } a = \frac{p}{q} \text{ where } p, q \in \mathbb{Z}.$$

Let b be some irrational number.

$$\text{Let their sum} = S \text{ such that } S = a+b \text{ and } S = \frac{x}{y} \text{ where } x, y \in \mathbb{Z}.$$

$$\text{Then } S - a = b.$$

But from S2 we know that two rational numbers summed give a rational number.

But b is irrational!

So our claim is false.

6 Claim: $x^2 - y^2 = 1$ has positive integer solutions.

$$x^2 - y^2 = (x+y)(x-y).$$

Assume a solution exists such that $x^2 - y^2 = 1$ and $x, y > 0$, and $x, y \in \mathbb{Z}$.

$$(x+y)(x-y) = 1$$

1 only has one factor: itself.

$$\therefore x+y = x-y = 1$$

$$\text{But if } x+y = x-y:$$

$$y = -y$$

$$y = 0$$

$$\text{But } x, y > 0.$$

$\therefore$ no solutions!

So our claim is false.

7 Claim: if n^2 is a multiple of 3, n is not a multiple of 3.

$$\text{Let } n = 3k+1$$

$$\begin{aligned}\text{Then } n^2 &= 9k^2 + 6k + 1 \\ &= 3(3k^2 + 2k) + 1 : \text{ not a multiple of 3}\end{aligned}$$

$$\text{Let } n = 3k+2$$

$$\begin{aligned}\text{Then } n^2 &= 9k^2 + 12k + 4 \\ &= 3(3k^2 + 4k + 1) + 1 : \text{ not a multiple of 3}\end{aligned}$$

$$\text{Let } n = 3k+3$$

$$= 3(k+1)$$

$$= 3k$$

$$\begin{aligned}\text{Then } n^2 &= (3k)^2 = 9k^2 \\ &= 3(3k^2) : \text{ multiple of 3!}\end{aligned}$$

So for n^2 to be divisible by 3, $3|n$.
 $\therefore$ our original claim was false.

6 Claim: $\sqrt{3}$ is rational.

if $\sqrt{3} \in \mathbb{Q}$, let $\sqrt{3} = \frac{a}{b}$ where $a, b \in \mathbb{Z}$ & a, b are coprime.

$$\text{Then } 3 = \frac{a^2}{b^2}.$$

$$3b^2 = a^2.$$

Here, if b is even, then a is even and a and b are not coprime.

Conversely if b is odd, then a must be odd.

$$3(2m)^2 = a^2$$

$$3(4m^2 + 4m + 1) = a^2$$

$$3(\underbrace{2(2m^2 + 2m + 1)}_{\text{odd}}) = a^2$$

$\therefore a^2$ is odd.

$$\text{Then let } a = 2n+1 \quad (\text{odd number})$$

$$b = 2m+1 \quad (\text{odd number})$$

$$\therefore 3b^2 = 3(4m^2 + 4m + 1)$$

$$= 12m^2 + 12m + 3$$

$$a^2 = 4n^2 + 4n + 1$$

$$12m^2 + 12m + 3 = 4n^2 + 4n + 1$$

$2(6m^2 + 6m + 1) = 2(2n^2 + 2n)$, so a & b are not coprime! $\therefore$ contradiction, our original claim is false.

Exam Questions

1 Let $5k$ be the greatest multiple of 5.

But $5k+5 = 5(k+1)$, a multiple of 5

and $5k+5 > 5k$

($5 > 0$), so $5k$ is not the greatest multiple of 5!

$\therefore$ No greatest multiple of 5.

2 Let n^2 be an even number.

n can either be even or odd.

If n is odd:

$$\text{Let } n = 2k+1$$

$$\text{Then } n^2 = (2k+1)^2$$

$$= 4k^2 + 4k + 1$$

$$= 2(2k^2 + 2k) + 1 : \text{ an odd number!}$$

So if n is odd, then n^2 is odd.

Now let n be even.

$$\text{So } n = 2j.$$

$$\text{Then } (2j)^2 = 4j^2 = 2(2j^2), \text{ an even number!}$$

So if n is even, then n^2 is even.

Now n^2 can only be even or odd.

If n^2 is odd, n cannot be even, as if n is even then n^2 is even.

So n must be odd.

Conversely, if n^2 is even, then n cannot be odd, as if n is odd then n^2 is odd.

So n must be even!

$\therefore$ if n^2 is even, n is even.